

DNS in Depth & DNS Attacks

What DNS Does

DNS maps human names (`example.com`) to IP addresses. It's the phone book of the internet – and one of the most attacked services.

The Resolution Chain

Browser → Local resolver → Root server (.) → TLD server (.com) → Authoritative server (example.com)

Local resolver (your router/ISP/DoH) asks about example.com

Root servers point to .com TLD

TLD servers point to the authoritative server

Authoritative server answers: "example.com = 93.184.216.34"

Record Types

Type	Meaning
A	IPv4 address
AAAA	IPv6 address
CNAME	Canonical (alias) name
MX	Mail exchange server
NS	Name server
TXT	Arbitrary text (SPF, DKIM)
PTR	Reverse (IP → name)

DNS Attacks

Cache Poisoning

The attacker injects a forged answer into a resolver's cache, sending victims to the attacker's server. Patch era: ID randomization + DNSSEC defend against this.

DNS Tunneling

Data encoded inside DNS queries/responses – a stealthy **covert channel**.
Example:

`secret1.attacker.com` query encodes "secret1"

The response dumps data back

Defenders must detect: suspicious domain patterns, excessive DNS traffic, unusual record types (`TXT`/large), high query volume without web use.

DNS as Data Exfil / C2 Channel

Malware uses DNS to:

Beacon ("I'm alive, here's my ID") in subdomains

Exfiltrate secrets (encode stolen data in hostname prefix)

Exfiltrate in TXT records encoded in hex/base64

Detection playbooks look for: high DNS volume, long random-looking prefixes (`.a.bleu.xyz` 60+ chars), low TTL domains used for C2.

Defense: DNSSEC & DNS Over HTTPS

DNSSEC – digitally signs DNS records so resolvers can verify authenticity

DNS over HTTPS (DoH) / DNS over TLS (DoT) – encrypts queries so ISPs/attackers can't see or tamper

Sinkholing – redirect malicious domains to a collector (used by blue teams & takedown operations)

Blocklists – deny known-malicious domains at the resolver

[← IP Addressing, Subnetting & ...](#)

[HTTP, HTTPS & TLS Deep Dive →](#)